

HerGuardian AI

Attribution

Credits for libraries, services, and reference sources used in building the HerGuardian AI platform.

1. Libraries and Frameworks

- Next.js
- React
- TypeScript
- Tailwind CSS
- shadcn-style UI components
- Lucide React Icons
- Framer Motion
- Recharts
- React Hook Form
- Zod
- Drizzle ORM
- PostgreSQL
- pg Node.js driver
- jsPDF
- html2canvas
- Axios
- Better Auth / Next Auth style authentication support

2. APIs and Services

- Vercel for deployment
- Neon PostgreSQL for database hosting
- Razorpay Checkout for payments
- Twilio for SMS, OTP, and call alerts
- Google Cloud OAuth for Google login
- OpenAI / Gemini / Puter.js-compatible AI APIs for chatbot support

3. Pretrained Models

1. A pretrained ML model, trained using the datasets listed in the Datasets section below (PHQ-9, NHANES-derived women's health data, BRFSS lifestyle data, and PCOS clinical data), is used to analyze health scores and mental wellness scores and generate accurate, personalized suggestions or possibilities of issues for users.
2. The chatbot uses externally hosted AI models provided through the configured OpenAI API to generate conversational responses and health guidance.

4. Datasets

The following public datasets were used to help train and calibrate the platform's mental health scoring, mood assessment, and symptom-screening logic:

- PHQ-9 (Patient Health Questionnaire-9) — the standard 9-item depression screening instrument, developed by Drs. Robert L. Spitzer, Janet B.W. Williams, Kurt Kroenke and colleagues, used as the scoring framework for mood and depressive-symptom assessment
- NHANES-derived Women's Health Indicators dataset — demographic, dietary, examination, laboratory, medication, and questionnaire data (including the PHQ-9 item set and reproductive health / menopause-related questions), used for symptom and indicator correlation
- CDC BRFSS (Behavioral Risk Factor Surveillance System) 2024 lifestyle codebook — used as a reference for lifestyle and behavioral risk factor variables
- PCOS (Polycystic Ovary Syndrome) clinical dataset — physical, hormonal, and lifestyle indicators used for PCOS-related symptom screening

All datasets used are publicly available research or survey data. No private or identifiable patient data is included in the repository.

The project also uses user-provided inputs for assessments, check-ins, logs, and reports.

5. Health Information Sources

Educational content was structured using publicly available health resources such as:

- WHO
- Mayo Clinic
- Cleveland Clinic
- NHS
- MedlinePlus
- NIMH
- WomensHealth.gov
- Johns Hopkins Medicine
- NCBI / PubMed Central

6. Pre-existing Work

No pre-existing proprietary codebase, template, or previously built project was reused for HerGuardian AI. The application was developed from scratch, using standard open-source libraries, frameworks, and publicly available API services and datasets, all listed above, as building blocks.

All custom application flows, UI integration, health-scoring and mental wellness-scoring logic, and feature combinations were independently created for HerGuardian AI.

This attribution list reflects the tools, services, and references used as of the current version of HerGuardian AI and may be updated as the platform evolves.